MIL-C-46275A(Ord)
25 June 1962

SUPERSEDING
MIL-C-46275(Ord)
30 August 1960

MILITARY SPECIFICATION
FOR
CASE, CARTRIDGE, 105MM, M150

1. SCOPE

1.1 This specification covers Case, Cartridge, 105MM, M150 for Cartridge HEP-T, M393 for M68 Gun.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitations for bids form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-P-10025	-	Packing and Marking for Domestic Shipment of Inert Ammunition Components, General Specification for
MIL-I-45208	-	Inspection Requirements, General Specification for

STANDARDS

Federal

Federal Test Method Standard No. 151 - Metals; Test Methods

FSC 1315

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Military
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109 - Inspection Terms and Definitions
MIL-STD-639 - Surface Roughness Values Preferred for Army Ordnance Materiel

DRAWINGS

Ordnance
F8597493 - Case, Cartridge, 105MM, M150
F7548320 - Cartons, Ammunition for Interplant Shipment of Brass Cartridge Cases, Empty, 75MM to 155MM, Inclusive
IEL 8597493 - Inspection Equipment List for Case, Cartridge, 105MM, M150

PUBLICATIONS

ORD-M608-11 - Ordnance Inspection Handbook - Procedures and Tables for Continuous Sampling by Attributes

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 General.- The cartridge case shall comply with the requirements of the applicable drawings, referenced specifications and the following:

3.2 Initial production sample.- Unless otherwise directed by the contracting officer, an initial production sample shall be required.

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3.3 Material.- The contractor shall be responsible for selecting from the classification list contained in the raw material specification, the surface and physical condition of the raw material intended for use in manufacture.

3.4 Annealing.- Annealing temperature shall not exceed 1200 degrees Fahrenheit (F).

3.5 Stress relief.- After the last cold forming operation, the cartridge case shall be given a stress relief treatment of approximately 500 degrees Fahrenheit (F) for one hour.

3.6 Mouth anneal.- After the final forming operation, the cartridge case shall be annealed at the mouth for a distance of 4 inches minimum 4 1/2 inches maximum.

3.7 Mercurous nitrate.- The cartridge case when fitted with a steel mouth plug simulating the maximum dimension of the 105mm, M393 projectile, shall show no evidence of cracking after immersion in mercurous nitrate.

3.8 Maximum profile.- The cartridge case shall freely enter the Ordnance certified maximum profile gage in such a manner as to permit the slide to close without interference.

3.9 Mechanical properties.-

3.9.1 Hardness.- The metal .120 inch minimum, from the surface around the primer hole shall have a Rockwell hardness value of not less than B65.

3.9.2 Tensile strength.- Specimens taken from sample cartridge cases in accordance with Figure I shall comply with the requirements of Table I. (See 6.1)

Table I

	Pounds Per Square Inch (PSI)
Specimen A	44,000 Minimum - 54,000 Maximum
Specimen B	50,000 Minimum
Specimen C	65,000 Minimum
Specimen D	65,000 Minimum
Specimen E	57,000 Minimum

Elongation shall be recorded for information only.

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3.10 Functional.- The cartridge case shall be subjected to a functional test in the following phases:

a. Excess Pressure Phase
b. Excess, Excess Pressure Phase (for information only)

3.10.1 Obturation.- The cartridge case shall effectively seal the breech of the weapon against leakage of hot gases generated during firing.

3.10.2 Soundness.- After firing, the cartridge case shall be free of cracks, except as permitted by Table III.

3.10.3 Ejection.- When fired, the cartridge case shall be ejected by the service extractors when the weapon is operated in the normal manner. (see 6.3).

3.10.4 Primer retention.- The cartridge case shall retain the primer during firing and extraction.

3.11 Workmanship.- The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.11.1 Metal defects.- The cartridge case shall be free from cracks, blisters, seams, laminations, deep scratches, and porosity.

3.11.1.1 Folds and cold shuts.- The cartridge case shall contain no folds or cold shuts in the radius area extending beyond the tangents from the base and sidewall, however, at no time will any fold or cold shut exceed .035 maximum length. See Figure II.

3.11.2 Surface finish.- The requirements for surface finish are as detailed on the applicable drawing.

3.11.3 Threads.- Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawing and provisions of Specification MIL-A-2550.

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3.11.4 Burr.- No cartridge case shall have a burr which might interfere with the assembly of the primer or projectile, or which might be injurious to personnel handling the cartridge case.

3.11.5 Foreign matter.- No cartridge case shall contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection terms and definitions.- Reference shall be made to Standard MIL-STD-109 to define inspection terms used.

4.1.2 Inspection.- Inspection shall be in accordance with Specification MIL-A-2550 and MIL-I-45208 except as specified herein.

4.2 Initial production sample.- Prior to initiation of regular production, the contractor shall submit an initial production sample to a Government approved facility for evaluation. The sample shall consist of seventy (70) cartridge cases for performance of examination and tests as specified in 4.3. Test procedures shall be in accordance with 4.4 except as provided in Specification MIL-I-45208. The samples shall be subjected to the tests listed in 4.3.3. The Government will perform ballistic tests on 60 of the above cartridge cases in accordance with 4.3.3 to determine compliance with 3.10. The sample shall be manufactured in the same manner using the same materials, equipment, processes and procedures as will be used in regular production. All parts and materials, including packaging and packing must be the same as will be used for regular production, and shall be obtained from the same source of supply. The contractor shall inspect the sample in accordance with his inspection system and other inspection requirements of Specification MIL-I-45208. After inspection and provisional acceptance at source, the initial production sample shall be inspected at the Government approved

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facility for all requirements of the drawings and specifications. The Government reserves the right to require new initial production samples until such time as an acceptable sample is submitted. The 60 cartridge cases to be used in ballistic testing shall be forwarded to a designated proving ground by the contractor.

4.3 Inspection provisions.-

4.3.1 Lot.-

4.3.1.1 Submittal of product.- The contractor shall submit the product in accordance with Standard MIL-STD-105 or Handbook ORD-M608-11 as applicable.

4.3.1.2 Size of lot.- For the purposes of identification and shipment, the lot size shall be as agreed upon between the contractor and the contracting officer, provided the homogeneity of the inspection lot has been retained.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plans contained in MIL-STD-105 or ORD-M608-11 as applicable. An AQL of 0.25 percent for each major defect and an AQL of 0.40 percent for each minor defect shall be used. All nonconforming material shall be rejected. Four power (4X) magnification may be used in performing visual examination for such defects as burr and foreign matter. The Ordnance Standard Finish blocks prescribed by Standard MIL-STD-639 shall be used as a basis of comparison for surface roughness determination.

4.3.2.1 Maximum profile gage.- Each cartridge case shall be gaged in an Ordnance certified maximum profile gage. Failure of any cartridge case to comply with requirements of 3.8 shall be cause for rejection of the cartridge case.

4.3.2.2 Classification of defects.-

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4.3.2.2.1 Case, Cartridge, Drawing F8597493

	Defect	Method of Inspection
MAJOR		
1.	Pitch diameter of primer thread, max.	Gage
2.	Minor diameter of primer thread, max.	Gage
3.	Concentricity of primer hole thread with body	Gage
4.	Concentricity of primer hole counterbore with thread	Gage
5.	Diameter of flange, min.	Gage
6.	Diameter of mouth	Gage
7.	Diameter of bead	Gage
8.	Location of bead	Gage
9.	Length of mouth diameter, min.	Gage
10.	Thickness of flange, min.	Gage
11.	Wall thickness at datum near head	Gage
12.	Wall thickness at datum near mouth	Gage
13.	Wall thickness at mouth	Gage
14.	Thickness of head, min.	Gage
15.	Depth of primer hole counterbore	Gage
16.	Metal defective	Visual
17.	Contour of inside of head and sidewall not smooth	Visual
MINOR		
41.	Pitch diameter of primer thread, min.	Gage
42.	Diameter of primer hole counterbore	Gage
43.	Diameter of body under flange	Gage
44.	Length to first datum diameter from head end, min.	Gage
45.	Length to second datum diameter from head end, min.	Gage
46.	Length to third datum diameter from head end min.	Gage

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Defect	Method of Inspection
MINOR (Cont'd)	
47. Length to fourth datum diameter from head end, min.	Gage
48. Total length	Gage
49. Chamfer on head missing or incorrect	Visual
50. Surface finish improper	Visual
51. Burr	Visual
52. Marking missing, incorrect or illegible	Visual
53. Foreign matter	Visual

4.3.2.2.2 Packed Carton, Prior to sealing, Drawing F7548320.

MAJOR	
None	
MINOR	
41. Support missing or damaged	Visual
42. Separator missing or damaged	Visual
43. Improper assembly	Visual
44. Incorrect number in carton	Visual

4.3.2.2.3 Packed Carton, After sealing, Drawing F7548320.

MAJOR	
None	
MINOR	
41. Carton damaged, cut or gouged through all layers	Visual
42. Carton not completely sealed	Visual
43. Contents loose	Manual/Visual
44. Marking missing, incorrect or illegible	Visual

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4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4 except as provided in Specification MIL-I-45208.

4.3.3.1 Mercurous nitrate.- Five cartridge cases from the initial production sample and five cartridge cases from each production lot shall be subjected to mercurous nitrate immersion. Failure of one cartridge case to comply with the requirements of 3.7 shall be cause for rejection of the sample or lot.

4.3.3.2 Mechanical properties.-

4.3.3.2.1 Hardness.- Five cartridge cases from the initial production sample and five cartridge cases from each production lot shall be subjected to this test. Failure of one cartridge case to comply with the requirements of 3.9.1 shall be cause for rejection of the sample or lot.

4.3.3.2.2 Tensile strength.- Five cartridge cases from the initial production sample and five cartridge cases from each production lot shall be subjected to this test. Failure of one cartridge case to comply with the requirements of 3.9.2 shall be cause for rejection of the sample or lot.

4.3.3.2.3 Elongation.- The percentage of elongation of each specimen used for tensile strength test shall be measured and recorded for information only.

4.3.3.2.4 Macro examination.- Five cartridge cases from the initial production sample and five cartridge cases from each production lot shall be examined for folds and cold shuts. Failure of one cartridge case to comply with the requirements of 3.11.1.1 will be cause for rejection of the sample or lot.

4.3.3.3 Functional.- The following functional tests will be performed by the Government.

4.3.3.3.1 Initial production sample.- The quantities listed in Table II will be ballistically tested in the phases indicated and the results of the test will be evaluated in accordance with criteria set forth in Table III except that no retest will be permitted.

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Table II

Test Phase	Quantity to be Tested
Excess Pressure Phase	30
Excess, Excess Pressure Phase (for information only)	30

4.3.3.3.2 Production lot sample.- Thirty cartridge cases from each production lot of 25,000 or less will be ballistically tested in the excess pressure phase. Lots which exceed 25,000 cartridge cases, one additional cartridge case will be tested for every one thousand cases (or fraction thereof) above 25,000. Acceptance and rejection criteria shall be in accordance with Table III. The lot shall be considered eligible for retest using double the number of samples provided that after firing, no cartridge case contains crack(s) or split(s) within one caliber of the head, and the rejection criteria of Table III has not been attained.

Table III

Defect Characteristic	First Sample Accept	First Sample Retest	Cumulative sample Accept	Cumulative sample Retest
1. Obturation	0	1	1	2
2. Splits or cracks within 1 caliber of head	0	0	0	1
3. Splits or cracks in sidewall more than 1 caliber from head	1	2	2	3
4. Ejection	0	1	1	2
5. Primer retention	0	1	1	2

4.3.4 Process control.- Monitoring at unscheduled intervals shall be conducted for the following process operations to determine compliance with following requirements.

Anneal - 3.4
Stress Relief - 3.5
Mouth Anneal - 3.6

Failure to comply with the requirements shall be cause for rejection of all units processed since the last monitored approval of the process.

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4.3.5 Inspection equipment.- The examination and tests shall be made using equipment listed on IEL-8597493, except as provided in Specification MIL-I-45208.

4.4 Test methods and procedures.-

4.4.1 Mercurous nitrate.- The cartridge case shall be tested in accordance with the procedures prescribed by Federal Test Method Standard No. 151. The cartridge case may be pickled for 30 seconds in a forty percent solution (specific gravity 1.25) of commercial nitric acid in lieu of the sulfuric acid solution specified.

4.4.2 Hardness.- The cartridge case shall be sectionalized through the center of the head. Three (3) or more hardness readings each shall be taken equally spaced through the head on both sides of the primer hole with the center line of the hardness indents located 0.100 inch from the major diameter of the primer hole thread. The hardness readings shall be made in accordance with the procedures prescribed by Federal Test Method Standard No. 151.

4.4.3 Tensile strength and elongation.- The test specimens shall be taken from the cartridge case in accordance with the standard positions and size of tensile specimen chart, Figure I. The tensile strength and elongation shall be determined in accordance with the procedures for these tests prescribed by Federal Test Method Standard No. 151.

4.4.4 Macro-examination.- The cartridge case shall be sectionalized through the center of the head in accordance with Figure II, polished and examined for folds and cold shuts in the radius areas of the head in accordance with the procedures prescribed by Federal Test Method Standard No. 151. One hundred (100X) magnification minimum shall be used for this examination.

4.4.5 Functional.-

4.4.5.1 Initial production sample.-

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4.4.5.1.1 Excess pressure phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. Complete round test components simulating the standard components may be used in lieu of the standard components. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees F to plus 75 degrees F and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.5.1.2 Excess, excess pressure phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. Complete round test components simulating the standard components may be used in lieu of the standard components. The propellant charge will be such as to produce the maximum chamber pressure permitted by the proving ground. The complete round will be temperature conditioned at plus 65 degrees F to plus 75 degrees F and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

4.4.5.2 Production sample.-

4.4.5.2.1 Excess pressure phase.- The cartridge case will be assembled into a complete round in the same manner as required for service firing. Complete round test components simulating the standard components may be used in lieu of the standard components. The propellant charge will be such as to produce the chamber pressure specified on the applicable drawing. The complete round will be temperature conditioned at plus 65 degrees F to plus 75 degrees F and fired in a serviceable weapon for which the round is standard. The weapon will be operated in the service manner.

5. PREPARATION FOR DELIVERY

5.1 Packing.-

5.1.1 Level A - Not applicable.

5.1.2 Level B - Not applicable.

5.1.3 Level C - Cartridge cases shall be packed in accordance with requirements of Drawing F-7548320.

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5.2 Labeling and marking.- Labeling and marking for shipment shall be in accordance with MIL-P-10025.

5.3 Data card.- Data card information shall be as specified in Specification MIL-A-2550.

6. NOTES

6.1 It is not implied that cartridge cases that contain minimum tensile strength values for all specimens will function satisfactorily in ballistic test.

6.2 The term "extraction" shall mean that the fired cartridge case is loosened and pulled to the rear of the weapon sufficiently to permit removal by hand.

6.3 The term "ejected" shall mean that the cartridge case is thrown free of the breech of the weapon to permit loading of the subsequent round.

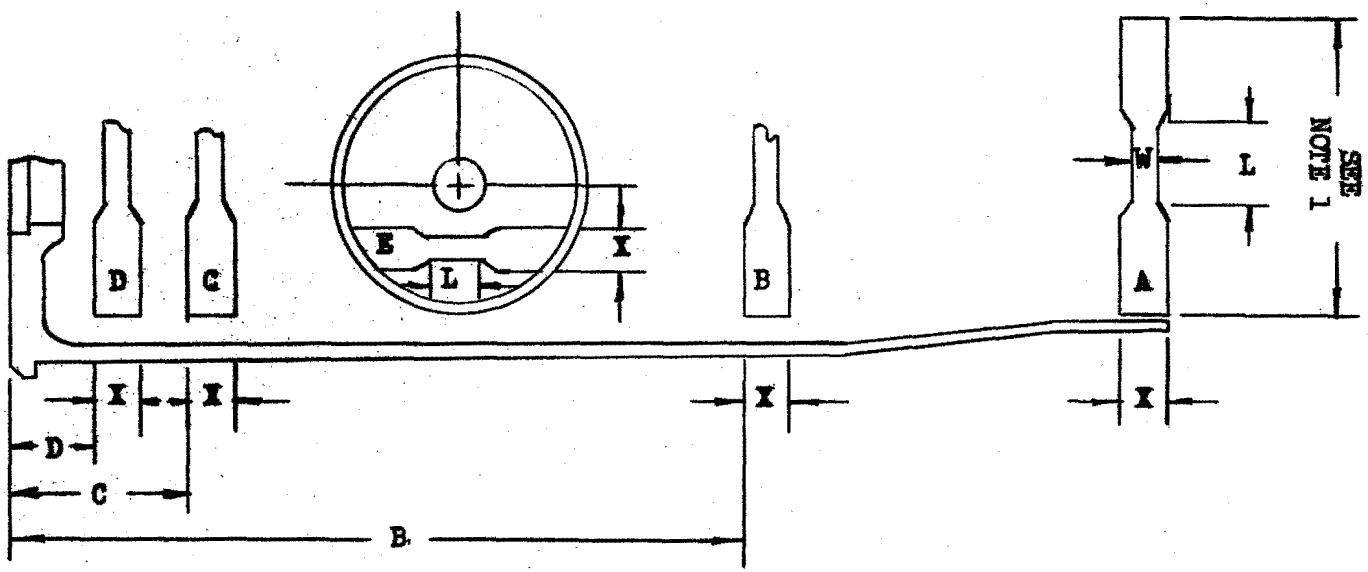
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STANDARD POSITIONS AND SIZES OF TENSILE SPECIMENS FOR CASE, CARTRIDGE, 105MM, M150



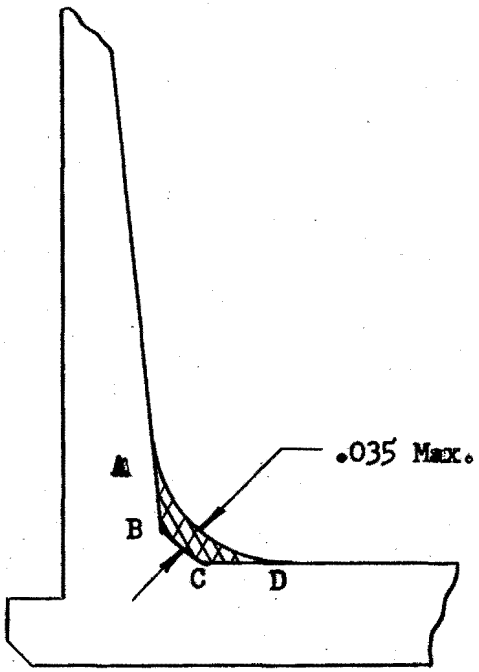
SPECIMEN	X	W	L	LOCATION FROM BASE
A	1.25	1.0	2.0	At Mouth
B	1.25	1.0	2.0	13.50
C	1.25	1.0	2.0	2.75
D	1.25	1.0	2.0	0.75
E	0.4	0.3	1.0	In Base

NOTES:

1. Overall length of specimen to suit fixture in testing machine.
2. Specimen E to be machined so as to eliminate effects of stamping on test.
3. Specimens C and D to be machined to uniform thickness.
4. Centerline of specimen E to be 1.75 inches from centerline of cartridge case.

Figure I

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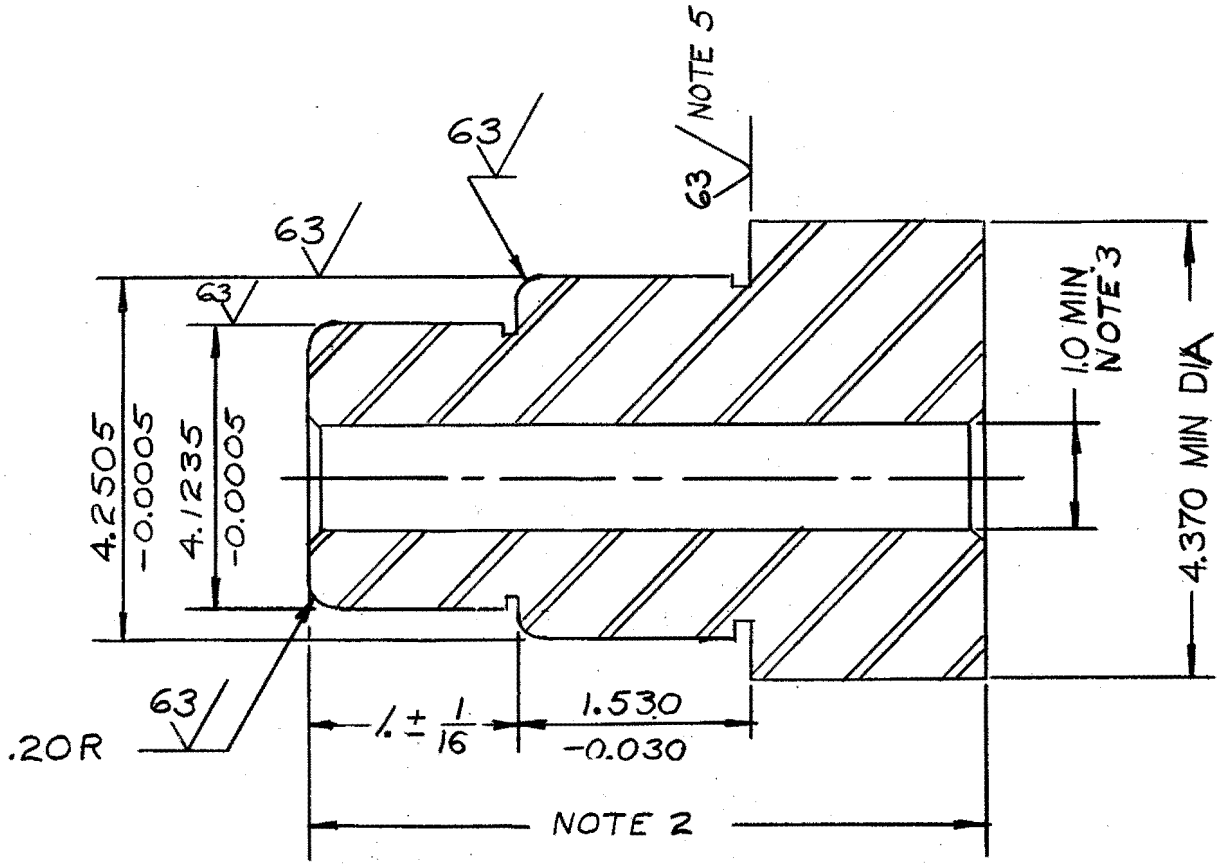


MACRO EXAMINATION

Folds or Cold Shuts shall not extend beyond shaded area inclosed by tangent AB, tangent CD and Arc BC.

Figure II

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NOTES:

1 - Material:- Steel
2 - Overall length of specimen to suit testing procedure
3 - Centers permitted; hole may be threaded, for extracting plug
4 - 250/ all over except as noted
5 - Cartridge case to be seated against this surface

MOUTH PLUG FOR USE IN MERCUROUS NITRATE TEST

Figure III

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